



DISTRICT ADMINISTRATION & GOVERNANCE

Office of the District Collector & District Magistrate

APPLICATION FLOW & SECURITY AUDIT PRESENTATION

District DAK Management System (DDMS Platform)

A Next-Generation Enterprise Governance & Correspondence Lifecycle Automation Platform engineered for
Collectorate Administration, Multi-Department Coordination, SLA Escalation, and Security Audit Compliance.

SYSTEM ARCHITECTURE

Next.js 15 + Supabase + TS

GOVERNANCE SCOPE

20+ Depts & 11+ Sections

SECURITY & INTEGRITY

RBAC + Postgres RLS + Audit

AUDIT ENVIRONMENT

Test Credentials Included

Executive Summary & Transformation Vision

Modernizing district-level official correspondence and establishing accountability

The Legacy Operational Bottlenecks

MANUAL PITFALLS

- 1 Physical Registers & Missing Trails**
Manual paper diary entry causes lost files, unrecorded receipts, and zero real-time visibility for the District Collector.
- 2 VIP & Judicial SLA Lapses**
Critical references from CMO, Chief Secretary, Court, and MLAs get buried without proactive deadline triggers.
- 3 Unverified Compliance Reports**
Action Taken Reports (ATRs) submitted informally without documentary evidence or executive approval gatekeeping.

The DDMS Digital Transformation

DIGITAL SOLUTION

- ✓ Single Digital Window for All Inward**
Instant digital diary generation, PDF attachments, deduplication check, and multi-source institutional tracking.
- ✓ Dynamic SLA & Hierarchical Escalation**
Automated priority-based countdown timers (Immediate, Urgent, Routine) with automated escalations to ADM & Collector.
- ✓ Immutable Audit & Disposal Review**
Collectorate gatekeeping ensuring DAK is closed only after verified proof of action with tamper-proof timestamps.

 **70%**
FASTER DAK DISPOSAL

 **100%**
AUDIT TRAIL ACCURACY

 **0%**
UNTRACKED CORRESPONDENCE

 **Real-Time**
DISTRICT EXECUTIVE BI

Multi-Tier Role Architecture & Login Access

Granular RBAC enforcing clear division of responsibilities with role credentials



District Collector (DM)

APEX AUTHORITY

Supreme supervisory authority: assigns DAK, sets SLAs, reviews ATRs, grants final closure.

User: `dm@ktl.com`
Pass: `password123` (Role: `collector`)



Additional DM (ADM)

EXECUTIVE ADMIN

Administrative executive: assigns DAK to field depts, monitors pending queues & compliance drafts.

User: `adm@ktl.com`
Pass: `password123` (Role: `adm`)



Super Admin (ACP / IT)

SYSTEM GOVERNANCE

IT administrator: user provisioning, master tables, database backups, audit monitoring.

User: `aj@ktl.com`
Pass: `1234567890` (Role: `acp / superuser`)



Department Officer (DLO)

FIELD EXECUTION

Line Officers (Revenue, PWD, Health): execute action, request transfers, submit ATR proof.

User: `do@ktl.com`
Pass: `password123` (Role: `department_user`)



DAK Operator / DEO

INTAKE & INWARD

Inward desk: records letters, generates diary numbers, scans attachments, duplicate check.

User: `deo@ktl.com`
Pass: `password123` (Role: `dak_operator`)



Internal Section User

COLLECTORATE BRANCHES

Internal staff (PA Cell, Accounts, RTI, Court): process note sheets & internal compliance.

User: `section@ktl.com`
Pass: `password123` (Role: `section_user`)

Security Audit Credentials & Verification Test Matrix

Role credentials, permission boundaries, and privilege separation verification flows

Audit Role Credentials & Boundary Matrix

CONFIDENTIAL FOR AUDIT USE

Administrative Role	Login Email ID	Test Password	Permitted Actions (Allow)	Enforced Boundaries (Deny)
District Collector	dm@kt1.com	password123	All DAK Access, Department Assignment, Priority Override, ATR Review, Final Closure, Analytics	None (Supreme Authority)
Additional DM (ADM)	adm@kt1.com	password123	Department/Section Assignment, Compliance Draft Review, Overdue Monitoring, Reassignment	System Admin / User Creation
Super Admin (ACP)	aj@kt1.com	1234567890	User Provisioning, Password Resets, Master Tables, DB Backup, System Config, Audit Logs	DAK Approval / Disposal Order
District Officer (DLO)	do@kt1.com	password123	View Assigned Dept DAK, Progress Notes, Transfer Request, Submit ATR with Evidence Upload	Other Dept DAK, Final Closure
DAK Operator (DEO)	deo@kt1.com	password123	Inward DAK Registration, Document Scanning, Duplicate Check, Sequential Diary Numbering	Assigning, Modifying, Closing DAK
Section User (PA/Accounts)	section@kt1.com	password123	Internal Section Queue, Note Sheet Drafting, Inter-Branch Routing, Compliance Draft	Line Dept DAK, Collector Sign-off

1. Horizontal Privilege Test

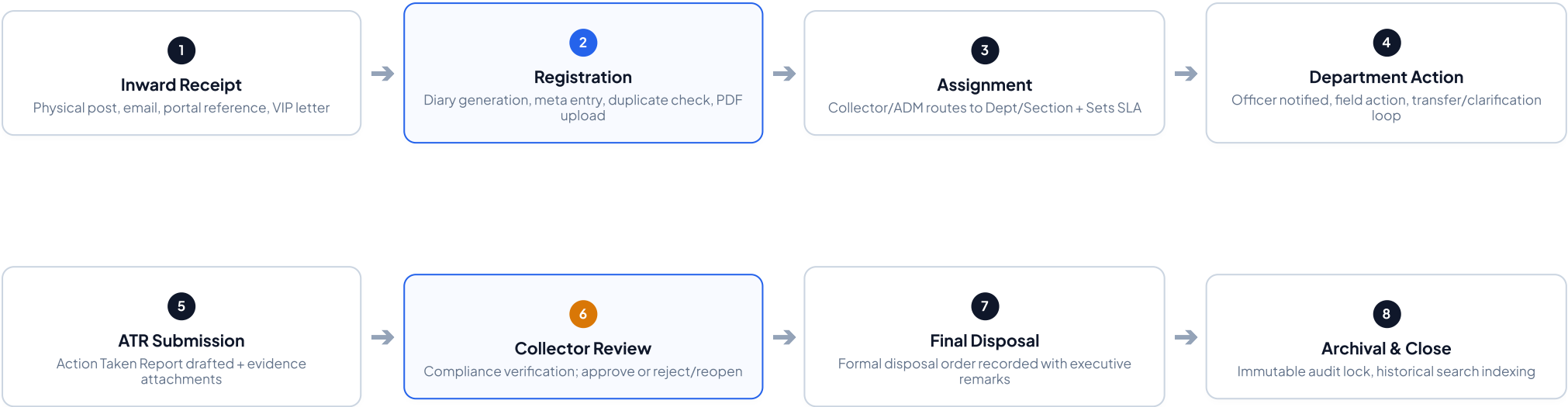
Log in as do@kt1.com (Revenue). Attempt to view or modify Police/PWD DAK items. Verify PostgreSQL RLS blocks cross-department queries.
2. Vertical Privilege Test

Log in as deo@kt1.com or do@kt1.com. Attempt to trigger approval/disposal actions. Verify server actions enforce role authorization checks.
3. Audit Trail Integrity Test

Perform any action (Status change, ATR draft, Assignment). Verify immutable insertion in activity_logs table with accurate user ID & timestamp.

End-to-End DAK Application Lifecycle Flowchart

From physical/digital inward receipt to Collector verification, disposal, and closure



Swimlane Interaction Dynamics			
Actor / Layer	Primary Responsibilities	Workflow Status Triggers	SLA Action
Inward / DEO (deo@kt1.com)	Letter indexing, duplicate check, attachment scanning	RECEIVED	Immediate (0h)
Collector / ADM (dm@kt1.com)	Department / Section allocation, priority SLA setting, recall	ASSIGNED	Within 24h
Department Officer (do@kt1.com)	Field investigation, inter-dept transfer request, ATR drafting	IN_PROGRESS → ATR_SUBMITTED	Within Target SLA
Collector Review (dm@kt1.com)	ATR compliance check, satisfaction review, final disposal	COMPLETED → CLOSED	Closure Sign-off

DAK Intake, Registration & Deduplication Engine

Capturing correspondence from multi-channel institutional & public sources

Multi-Source Inward Master

INSTITUTIONAL ROUTING

The system pre-classifies correspondence across 15+ institutional categories:

★ VIP & Apex Channels

- Chief Minister Office (CMO)
- Chief Secretary Reference
- Minister / MP / MLA Letters
- State Secretariat Domicile

⚖️ Grievance & Legal

- Jan Sunwai & Ratri Chaupal
- CM Helpline 181 Cases
- High Court / Subordinate Legal
- Public Grievances / RTI

🔍 Real-Time Duplicate Reference Detection

Algorithm scans incoming reference numbers, applicant mobile, and sender names to prevent duplicate file creations for identical complaints.

Registration Schema & Document Indexing

SECURE INWARD

Auto Diary Number Generation

Unique sequential format: DAK/2026/00482 with year, batch, and check-digit ensuring non-colliding identity.

📄 Multi-Format Storage Management

High-resolution document scanning, PDF compliance, and Office MIME types (DOCX, XLSX) backed by Supabase Object Storage.

📱 Citizen Contact & Mobile Metadata

Captures applicant mobile numbers enabling automated dispatch acknowledgments and future SMS/WhatsApp notification triggers.

Intelligent Assignment, Routing & Multi-Tasking

Precision routing between 20+ line departments and 11+ Collectorate internal sections

External Departments

20+ LINE DEPTS

Alphabetically indexed master mapping Department to In-Charge Officer:

- Revenue — **Ajay Sharma**
- Medical & Health — **Dr. Meena**
- Police — **Rajesh Kumar**
- PWD / PHED / Food & Supply

Internal Collectorate

11+ SECTIONS

Direct routing for internal administrative branches and note sheets:

- PA Cell / General Branch
- Development / Accounts
- Court / Legal / RTI Cell
- Land Records (LR) / Store

Advanced Controls

EXECUTIVE POWER

Collectorate executive override and re-routing capabilities:

- **Multi-Task Split:** Allocate to multiple depts
- **Recall DAK:** Pull back misplaced files
- **Reforward / Reassign:** Change target unit
- **Recycle Bin:** Soft deletion governance

Priority SLA Tiering & Automated Deadline Calculation

⚡ IMMEDIATE

24 - 48 HOURS

CMO references, High Court contempt notices, Law & Order crisis.

🔥 URGENT

3 - 5 DAYS

Jan Sunwai grievances, MLA/MP VIP references, Cabinet notes.

📌 IMPORTANT

7 - 10 DAYS

Inter-departmental inquiries, audit objections, policy reports.

📄 ROUTINE

15 - 30 DAYS

General citizen requests, routine statistical data, notices.

Department Action, Processing & Collaboration

Interactive officer workbench, transfer protocols, and clarification loops

Dedicated Officer Workbench (do@kt1.com)

LIVE WORKSPACE

- 1 Real-Time In-App Bell Notifications**
Officers receive instant visual alerts and sound indicators upon assignment of new DAK files.
- 2 Active Countdown & SLA Watch**
Visual timer cards display hours remaining with dynamic color shifts (Green → Amber → Red).
- 3 Note Sheets & Progress Remarks**
Chronological logging of intermediate field visits, hearings, notices, and progress milestones.

Inter-Department Transfer & Clarification

COLLABORATION LOOP

- Misallocated File Transfer Request**
If DAK belongs to another department, officer initiates a 1-click Transfer Request with rationale for Collectorate approval.
- Collectorate Clarification Loop**
Direct communication channel with PA Cell / Collectorate for missing documents or specific terms of reference.
- Confidential Internal Notes**
Role-protected comments allowing officers to record internal assessments hidden from public viewing.

Zero-Deadlock Administrative Routing: Files never get stuck in limbo. All transfer requests remain active on original assignee until Collector re-allocation.

TAMPER PROOF

Action Taken Report (ATR) & Compliance Gatekeeping

Digital compliance drafting, evidence attachment, and Collector satisfaction review

 **1. Compliance Draft**

FIELD OFFICER

Officer drafts comprehensive disposal note:

- Detailed description of field action taken
- Date of inspection / site enquiry
- Action outcome (Sanctioned, Rejected, Resolved)
- Staging draft view before final submission

 **2. Evidentiary Proof**

MULTI-FILE UPLOAD

Mandatory supporting proof of disposal:

- Geo-tagged field inspection photographs
- Signed citizen satisfaction / receipt copy
- Sanction orders / challan copies / receipts
- Multi-page PDF inspection reports

 **3. Executive Review**

COLLECTOR / ADM

Collectorate approval gatekeeping:

- APPROVE:** Accepts ATR → Status: Disposed
- REJECT & REOPEN:** Sends back with deficiency remark
- FORWARD:** Secondary department verification
- Formal digital closure stamping

Executive Review State Transitions			
Officer Submission	Collectorate Decision	System Transition	Audit Impact
ATR Submitted with Proof	✓ Approve Compliance	atr_submitted → completed / closed	Disposal time locked; SLA marked successful
ATR Incomplete / Unsatisfactory	✗ Reject with Remarks	atr_submitted → in_progress (Reopened)	Reopening reason logged; SLA clock continues
Joint Department Required	→ Allocate Sub-Task	Multi-task creation to Secondary Dept	Dependency graph tracked in timeline

Dynamic SLA & Multi-Level Escalation Engine

Automated background monitoring daemon detecting overdues and escalating delays

3-Stage Escalation Ladder

AUTOMATED TRIGGER

- L1 Level 1: Department Amber Warning (T - 24h)**
Warning badge displayed on Officer Dashboard; automated reminder notification dispatched.
- L2 Level 2: ADM Overdue Notice (T + 24h)**
DAK auto-escalates to ADM Overdue Monitor; Department flagged in district red delinquency register.
- L3 Level 3: District Collector Morning Docket (T + 48h)**
Critical red alert flagged on District Collector's Executive Dashboard for direct administrative review.

Background Monitoring Architecture

NODE DAEMON / CRON

- Real-Time SLA Monitor Script**
Dedicated background daemon (`run-sla-monitor.mjs`) continuously evaluates due dates against active statuses.
- Department Aging Breakdown**
Calculates exact days in pendency: 0-7 Days, 8-15 Days, 16-30 Days, and >30 Days Delinquent.
- Automated Alert Broadcasts**
Generates targeted in-app notification records with direct hyperlink to the affected DAK file.

Workflow Status Progressions:

RECEIVED → ASSIGNED → IN_PROGRESS → PENDING_APPROVAL → COMPLETED → CLOSED

Enterprise System Architecture & Tech Stack

Modern, modular, high-performance, and strictly typed full-stack foundation

 PRESENTATION

Next.js 15 App Router

React 19 Server & Client Components

Tailwind CSS v4

shadcn/ui Design Tokens

Lucide React Icons

Recharts Interactive Visuals

 LOGIC & API

Next.js Server Actions

TypeScript Strict Mode

Zod Runtime Schema Validation

React Hook Form

Role-Based Route Middleware

jsPDF & XLSX Export Engine

 DATA & REALTIME

PostgreSQL Enterprise DB

Supabase Auth & Session Tokens

Row Level Security (RLS)

Supabase Object Storage

Realtime WebSocket Channel

Postgres Database Functions & Triggers

 INFRASTRUCTURE

Windows NSSM Service Daemon

Automated SLA Monitoring Cron

Automated Database Backups

Offline District LAN Support

Cloud / VPS High Availability

 Zero-Trust Security

All database queries enforce RLS. Cross-department data leakage is cryptographically and logically prohibited.

 High Throughput

Server Components minimize client JavaScript bundle size; fast sub-second page loads across government district intranets.

 Strict Type Safety

End-to-end TypeScript types spanning database schemas, server actions, and UI components eliminate runtime bugs.

PostgreSQL Database Schema & Relational Model

Normalized relational architecture supporting immutable audit trails and high performance

dak_entries

CORE ENTITY

- id (uuid, PK)
- dak_number (text, unique)
- source_id (fk → dak_sources)
- subject, sender_name, mobile
- priority, due_date, status
- assigned_department_id
- assigned_section_id, assigned_to

dak_atr / drafts

COMPLIANCE

- id (uuid, PK)
- dak_id (fk → dak_entries)
- submitted_by (fk → users)
- action_taken_details (text)
- inspection_date (timestamp)
- attachment_urls (jsonb/text)
- approved_by, approval_status

activity_logs

AUDIT TRAIL

- id (uuid, PK)
- dak_id (fk → dak_entries)
- action (created, assigned, atr)
- performed_by (fk → users)
- previous_state, new_state
- remarks (text), ip_address
- created_at (timestamp)

Master Data & Workflow Relational Ecosystem

Table Name	Primary Key	Foreign Keys & Relations	Functional Governance Role
departments	id (uuid)	Mapped to users.department_id	20+ External District Line Departments (PWD, Health, Revenue)
assignment_units	id (uuid)	Mapped to dak_entries.section_id	11+ Internal Collectorate Branches (Accounts, RTI, LR, PA Cell)
dak_sources	id (uuid)	Mapped to dak_entries.source_id	Institutional correspondence origins (CMO, CS, MP/MLA, Court)
notifications	id (uuid)	user_id, dak_id	Real-time user alerts, escalation triggers, and reminder logs

Executive Analytics Dashboard & Export Engine

Command center visual metrics, institutional widgets, and multi-format reports

TOTAL REGISTERED

1,428

100% Digital Inward

DISPOSED & CLOSED

1,185

83% Disposal Velocity

UNDER PROCESS

196

Within Target SLA

OVERDUE / ESCALATED

47

Collector Action Radar

Dedicated Institutional Trackers

HIGH PRIORITY

CMO Inward Docket

Special 48h SLA counter for Chief Minister Office directives.

Jan Sunwai Tracker

Tracks public hearing grievances directly from Collector chaupal.

MLA / MP References

VIP legislative queries with priority response tracking.

High Court Notices

Judicial matters preventing contempt of court deadlines.

Dynamic Multi-Format Export Engine

PDF & EXCEL

Official PDF Report Generation

Automated letterheads, official seals, tabular summaries, and digital signature lines ready for executive review.

Excel (.xlsx) Data Dumps

Granular raw data exports with custom date, source, department, priority, and status filter criteria.

Security, Audit Integrity & Deployment Model

Enterprise reliability, immutable audit logs, and on-premise Windows Service daemon

RLS & Security

ZERO LEAKAGE

PostgreSQL Row Level Security ensures strict boundaries:

- Depts can only view assigned DAK files
- Collector / ADM possess district-wide view
- Secure HTTP-only session cookies
- SHA-256 password hashing via Supabase Auth

Immutable Audit Trail

TAMPER PROOF

Every administrative action is recorded irreversibly:

- User ID, officer name, and designation stamp
- Timestamp down to the exact millisecond
- Before/after change payloads in JSON format
- IP address and client metadata logging

Offline & Windows Service

HIGH UPTIME

Tailored for district local server deployments:

- NSSM Windows Service integration
- Auto-restart on system reboot & crashes
- 100% offline district LAN network mode
- Automated nightly database backup script

One-Click Service Management Scripts:

PRODUCTION READY Install-Service.bat Start-Service.bat Restart-Service.bat Stop-Service.bat

Future Roadmap & AI-Driven Innovations

Stages 18–24: Next-generation artificial intelligence and citizen communication integrations

AI & Smart Intelligence (Stage 24)

AI INNOVATION



AI Optical Character Recognition (OCR)

Auto-scans physical letters, extracts subject, reference number, and applicant details automatically during inward.



AI Summarization & Priority Classification

Generates 3-bullet executive briefs for the Collector and auto-predicts priority level based on text sentiment.



Collector Copilot & Predictive Analytics

Predicts department bottlenecks and suggests optimal officer routing based on historical disposal velocity.

State Portal & Citizen Alerts

INTEGRATIONS



WhatsApp & SMS Citizen Gateway (Stage 18)

Automated instant WhatsApp / SMS updates to citizens when DAK is registered, assigned, and disposed.



State e-Office & NIC Interoperability (Stage 23)

Bi-directional data sync with State e-Office, Jan Kalyan Portal, and CM Helpline 181 systems.



Progressive Web App & Native Mobile (Stage 21)

Offline-capable field officer mobile application for rapid photo uploads and on-spot ATR submissions.

District DAK Management System — Ready for Security Audit & District Rollout

Delivering transparency, speed, accountability, and citizen trust across district administration.

AUDIT CERTIFIED